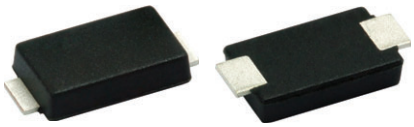
Surface-Mount TMBS® (Trench MOS Barrier Schottky) Rectifier

eSMP® Series



Top View Bottom View

SlimSMA (DO-221AC)

Cathode ○—▶|—○ Anode

FEATURES

- Very low profile - typical height of 0.95 mm
- Ideal for automated placement
- Trench MOS Schottky technology
- Low power losses, high efficiency
- Low forward voltage drop
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available
  - Automotive ordering code: base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE GRADE Available

RoHS COMPLIANT

HALOGEN FREE

LINKS TO ADDITIONAL RESOURCES

3D Models

TYPICAL APPLICATIONS

For use in high frequency inverters, freewheeling, DC/DC converters, and polarity protection in commercial, industrial, and automotive applications.

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	5.0 A
V_{RRM}	60 V
I_{FSM}	100 A
V_F at I_F = 5.0 A (125 °C)	0.52 V
T_J max.	175 °C
Package	SlimSMA (DO-221AC)
Circuit configuration	Single

MECHANICAL DATA

Case: SlimSMA (DO-221AC)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-M3 - halogen-free, RoHS-compliant
Base P/NHM3 - halogen-free, RoHS-compliant, and AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
M3 and HM3 suffix meets JESD 201 class 2 whisker test

Polarity: color band denotes cathode end

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)			
PARAMETER	**SYMBOL**	**VSSAF5M6**	**UNIT**
Device marking code		5M6	
Maximum repetitive peak reverse voltage	V_{RRM}	60	V
Maximum DC forward current	$I_{F(AV)}$ (1)	2.8	A
	$I_{F(AV)}$ (2)	5.0	
Peak forward surge current 10 ms single half sine-wave superimposed on rated load	I_{FSM}	100	A
Operating junction and storage temperature range	T_J, T_{STG}	-40 to +175	°C

Notes

(1) Free air, mounted on recommended copper pad area
(2) Mounted on 30 mm x 30 mm pad area

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Instantaneous forward voltage	I_F = 2.5 A	T_A = 25 °C	V_F (1)	0.52	-	V
	I_F = 5.0 A			0.58	0.66	
	I_F = 2.5 A	T_A = 125 °C		0.42	-	
	I_F = 5.0 A			0.52	0.6	
Reverse current	V_R = 60 V	T_A = 25 °C	I_R (2)	-	0.35	mA
		T_A = 125 °C		2	7	
Typical junction capacitance	4.0 V, 1 MHz		C_J	580	-	pF

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 40 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise specified)			
PARAMETER	**SYMBOL**	**VSSAF5M6**	**UNIT**
Typical thermal resistance	$R_{\theta JA}$ (1)(2)	115	°C/W
	$R_{\theta JM}$ (3)	12	

Notes

(1) Free air, mounted on recommended PCB, 2 oz. pad area; thermal resistance $R_{\theta JA}$ - junction to ambient, $R_{\theta JM}$ - junction to mount

(2) The heat generated must be less than thermal conductivity from junction-to-ambient: $dP_D/DT_J < 1/R_{\theta JA}$

(3) Mounted on 30 mm x 30 mm pad area

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PREFERRED PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
VSSAF5M6-M3/H	0.032	H	3500	7" diameter plastic tape and reel
VSSAF5M6-M3/I	0.032	I	14 000	13" diameter plastic tape and reel
VSSAF5M6HM3/H (1)	0.032	H	3500	7" diameter plastic tape and reel
VSSAF5M6HM3/I (1)	0.032	I	14 000	13" diameter plastic tape and reel

Note

(1) AEC-Q101 qualified

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

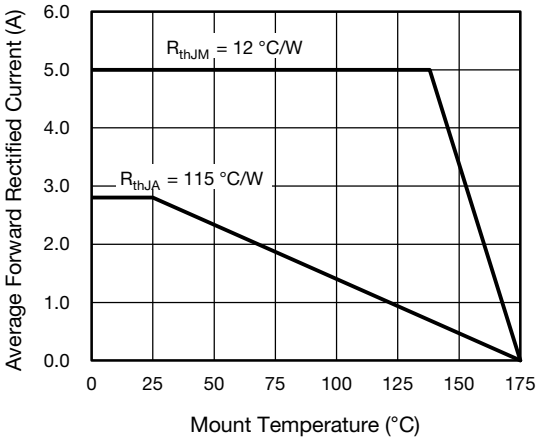


Fig. 1 - Maximum Forward Current Derating Curve

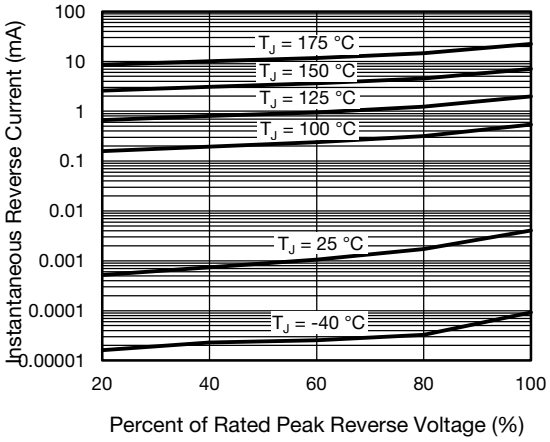


Fig. 4 - Typical Reverse Leakage Characteristics

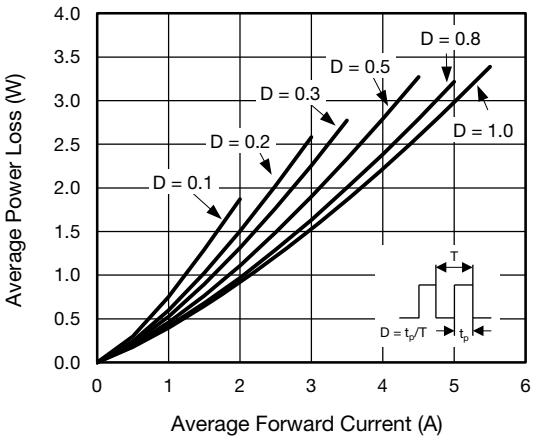


Fig. 2 - Forward Power Loss Characteristics

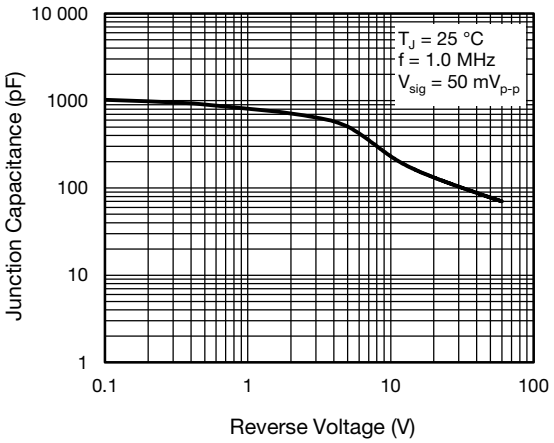


Fig. 5 - Typical Junction Capacitance

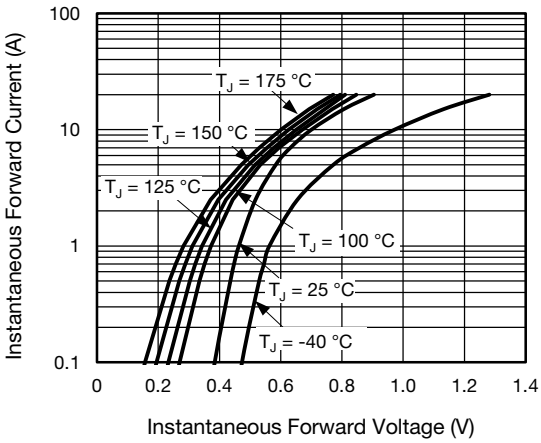


Fig. 3 - Typical Instantaneous Forward Characteristics

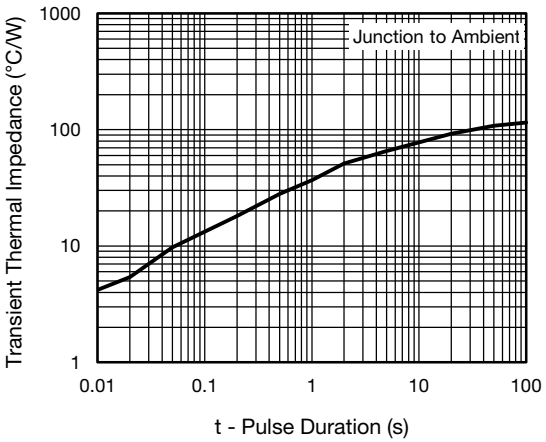


Fig. 6 - Typical Transient Thermal Impedance

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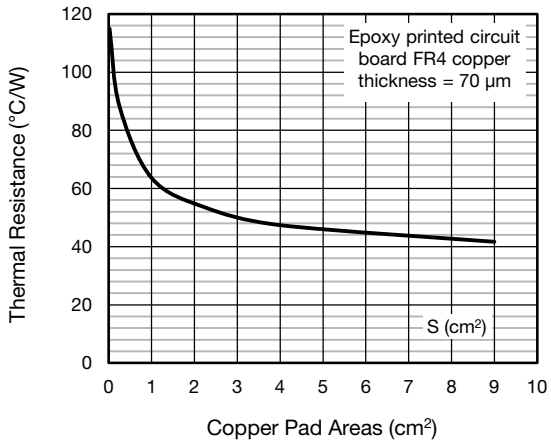
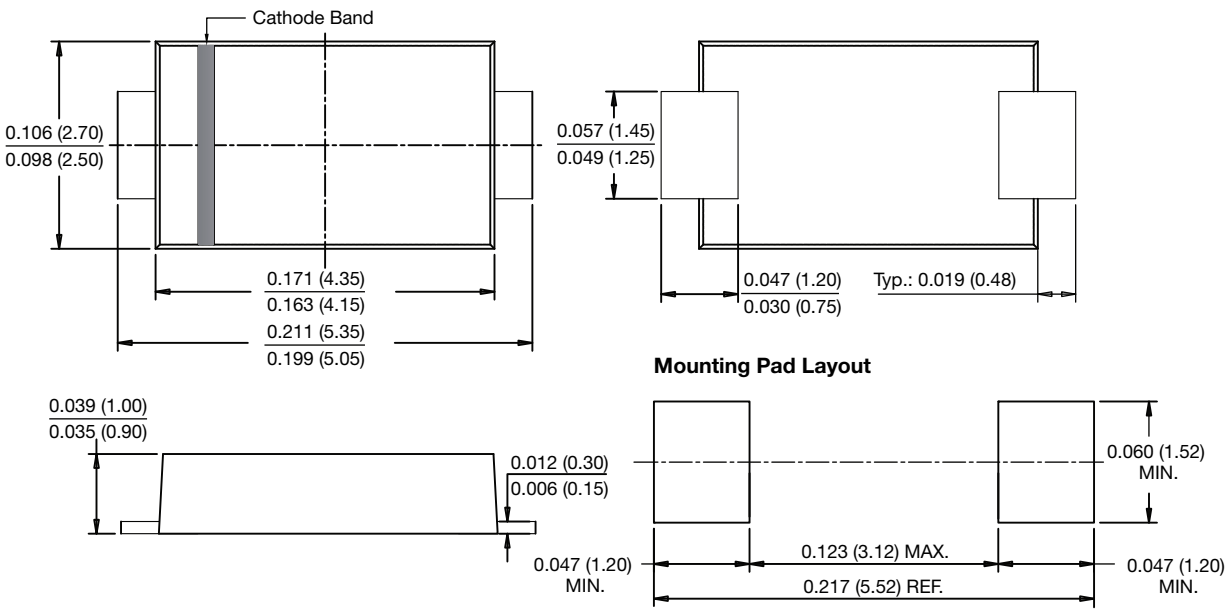


Fig. 7 - Thermal Resistance Junction to Ambient vs. Copper Pad Area

PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

SlimSMA (DO-221AC)



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